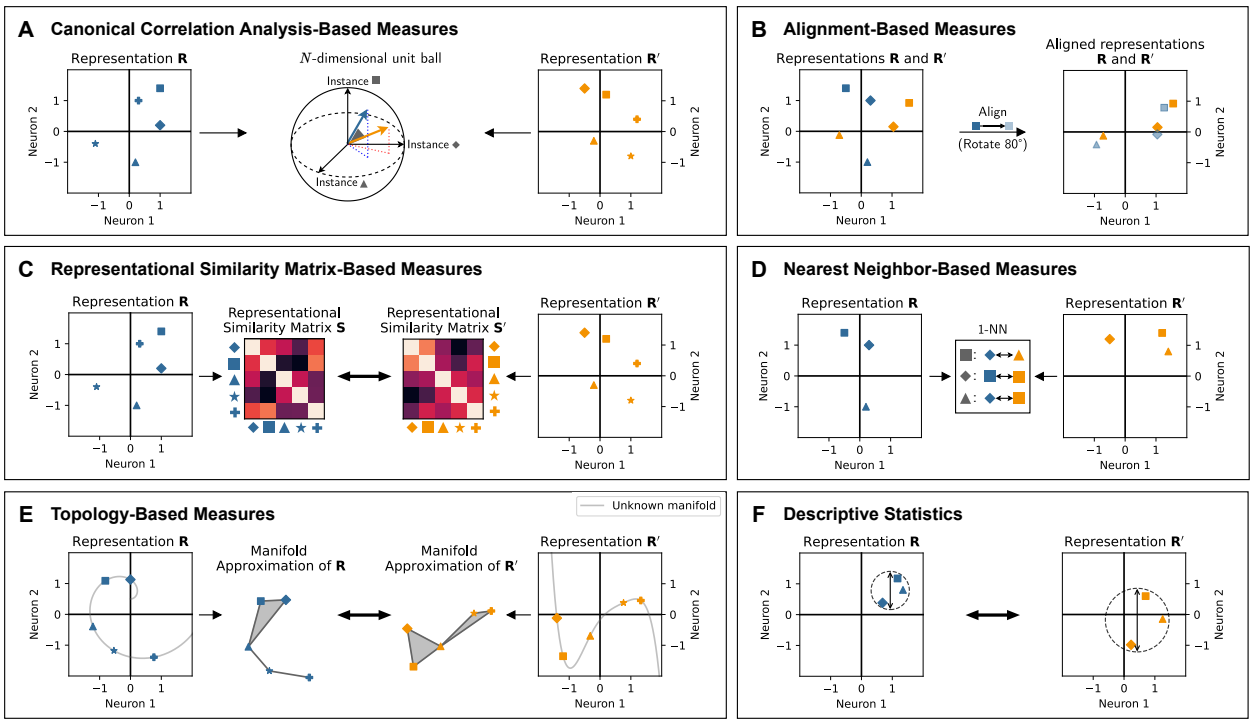

Figure 3: Types of representational similarity measures, illustrated with 2-dimensional representations. **A:** Representations of N instances are projected onto the N-dimensional unit ball, and similarity is then quantified based on their angle (their correlation). The illustration of the unit ball is not to scale, and only the first three dimensions are shown. **B:** Representations are aligned with each other, and similarity is computed after alignment. **C:** Similarity is based on comparing matrices of pairwise similarities within representations. **D:** Representations are compared based on similarity of their k nearest neighbors, here $k = 1$. **E:** Manifolds of the representations are approximated and compared. **F:** Statistics are computed individually for each representation (here: spread of instance representations) and then compared.

CCA is invariant to affine transformations [101]. If the representations $\boldsymbol{R}, \boldsymbol{R'}$ are equivalent, it holds that $\rho_i = 1$ for all $i \in \{1, \ldots, D\}$ and thus $m_{\text{CCA}}(\boldsymbol{R}, \boldsymbol{R'}) = 1$ and $m_{\text{CCA}^2}(\boldsymbol{R}, \boldsymbol{R'}) = 1$.

Other prominent aggregation schemes, though not applied for representational similarity, include the sum of the squared canonical correlations (also known as Pillai's trace [113]), Wilk's lambda statistic [154], and the Lawley-Hotelling trace [78, 59]. Several more aggregation methods can be applied, and there are numerous variants of CCA measures, including non-linear and multi-view ones—overviews on such variants are provided in the recent survey by Yang et al. [160] or the tutorial by Uurtio et al. [146]. In this work, however, we only consider those CCA-based measures that have been used to measure representational similarity of neural networks.

Singular Value CCA. Raghu et al. [114] argued that representations are noisy and that this noise should be removed before conducting CCA on the representations $\boldsymbol{R}, \boldsymbol{R'}$. Thus, they proposed the *Singular Value CCA (SVCCA)* approach, in which denoised representations are obtained by performing PCA on the representations. The number k of principal components that are kept is selected such that a fixed relative amount t of the variance in the data, usually 99 percent, is explained. Afterward, they use standard CCA on the denoised representations. Thus, letting $\widetilde{\boldsymbol{R}}, \widetilde{\boldsymbol{R}}'$ denote the denoised representations, the average canonical correlation is used as the final similarity measure:

$$m_{\text{SVCCA}}(\boldsymbol{R}, \boldsymbol{R'}) = m_{\text{CCA}}(\widetilde{\boldsymbol{R}}, \widetilde{\boldsymbol{R}}'). \tag{13}$$

Practically, the representations are also mean-centered before the PCA denoising. Unlike CCA, SVCCA is only invariant to orthogonal transformations, isotropic scaling and translation. SVCCA is bounded in the interval $[0, 1]$, with a score of one indicating perfectly similar representations.

To compute SVCCA efficiently for CNNs with many features, Raghu et al. [114] applied a Discrete Fourier Transform on each channel, yielding block-diagonal matrices for CCA computation, which eliminates unneeded operations.

Type	Measure	Invariances PT	OT	IS	ILT	TR	AT	Preprocessing	$D \neq D'$	Metric	Similarity ↑
Canonical Correlation Analysis	Mean Canonical Correlation [114]	✓	✓	✓	✓	✓	✓	CC	✓	✗	✓
	Mean Squared Canonical Correlation [73, 159]	✓	✓	✓	✓	✓	✓	CC	✓	✗	✓
	Singular Vector Canonical Correlation Analysis (SVCCA) [114]	✓	✓	✓	✗	✓	✗	CC	✓	✗	✓
	Projection-Weighted Canonical Correlation Analysis (PWCCA) [101]	✗	✗	✓	✗	✓	✗	CC	✓	✗	✓
Alignment	Orthogonal Procrustes [34, 155]	✓	✓*	✗	✗	✗	✗	✗	✗	✓	✗
	Angular Shape Metric [155]	✓	✓*	✓	✗	✗	✗	MN	✗	✓	✗
	Partial Whitening Shape Metric [155]	✓	✓	✓†	✓†	✓	✓†	✗	✗	✓	✗
	Soft Matching Distance [65]	✓	✗	✓	✗	✓	✗	CC, MN	✓	✓	✗
	Linear Regression [81, 73]	✓	✓	✓	✗	✓	✗	CC	✓	✗	✗
	Aligned Cosine Similarity [54]	✓	✓	✓	✗	✗	✗	✗	✗	✗	✓
	Correlation Match [81]	✓	✗	✓	✗	✓	✗	CC	✗	✗	✓
	Maximum Matching Similarity [152]	✓	✓	✓	✓	✗	✗	✗	✗	✗	✓
	ContraSim [116]	✓†	✓†	✓†	✓†	✓†	✓†	✗	✓	✗	✓
Representational Similarity Matrix	Norm of Representational Similarity Matrix Difference [125, 162]	✓‡	✓‡	✗‡	✗	✗‡	✗	✗	✓	✓‡	✗
	Representational Similarity Analysis (RSA) [74]	✓‡	✗‡	✓‡	✗	✓‡	✗	✗	✓	✗	✓‡
	Centered Kernel Alignment (CKA) [73]	✓	✓	✓	✗	✓	✗	CC	✓	✗	✓
	Distance Correlation (dCor) [139]	✓	✓	✗	✗	✓	✗	✗	✓	✗	✓
	Normalized Bures Similarity (NBS) [141]	✓	✓	✓	✗	✗	✗	✗	✓	✗	✓
	Eigenspace Overlap Score (EOS) [95]	✓	✓	✓	✓	✗	✗	✗	✓	✗	✓
	Unified Linear Probing (GULP) [17]	✓	✓	✓	✗†	✓	✗†	CC, RN	✓	✓	✗
	Riemmanian Distance [125]	✓	✓	✗	✗	✗	✗	✗	✓	✓	✗
	Relational Knowledge Loss [112]	✓	✓	✓†	✗	✓	✗	✗	✓	✗	✗
Neighbors	k-NN Jaccard Similarity [124, 149, 61, 52]	✓	✓	✓	✗	✗‡	✗	✗	✓	✗	✓
	Second-Order Cosine Similarity [53]	✓	✓	✓	✗	✗	✗	✗	✓	✗	✓
	Rank Similarity [149]	✓	✓	✓	✗	✗‡	✗	✗	✓	✗	✓
	Joint Rank and Jaccard Similarity [149]	✓	✓	✓	✗	✗‡	✗	✗	✓	✗	✓
Topology	Geometry Score (GS) [66]	✓	✓	✓	✗	✓	✗	✗	✓	✗	✗
	Multi-Scale Intrinsic Distance (IMD) [144]	✓	✓	✓	✗	✓	✗	✗	✓	✗	✗
	Representation Topology Divergence (RTD) [8]	✓	✓	✓	✗	✓	✗	✗	✓	✗	✗
Statistic	Intrinsic Dimension [21]	✓	✓	✓	✓	✓	✓	✗	✓	✗	§
	Magnitude [149]	✓	✓	✗	✗	✗	✗	✗	✓	✗	§
	Concentricity [149]	✓	✓	✓	✗	✗	✗	✗	✓	✗	§
	Uniformity [153]	✓	✓	✗	✗	✓	✗	✗	✓	✗	§
	Tolerance [150]	✓	✓	✓	✗	✗	✗	RN	✓	✗	§
	Instance-Graph Modularity [122, 87]	✓	✓	✓†	✗	✗	✗	✗	✓	✗	§
	Neuron-Graph Modularity [77]	✓	✗	✗	✗	✗	✗	✗	✓	✗	§

*: Subgroups possible. †: Varies based on hyperparameters. ‡: Similarity function dependent. §: Depends on comparison.

Table 1: Overview of representational similarity measures. The invariances are permutation (PT), orthogonal transformation (OT), isotropic scaling (IS), invertible linear transformation (ILT), translation (TR), and affine transformation (AT). We report invariances based on default hyperparameters and preprocessing as proposed by the authors. These may vary if different parameters or similarity functions are applied. Three kinds of preprocessing are commonly used: centering columns (CC), normalizing the matrix norm (MN), or normalizing row norms (RN). The column $D \neq D'$ indicates whether a measure requires the compared representations to have identical dimensionality. *Metric* indicates whether a similarity measure satisfies the criteria of a distance metric when representations have equal dimensionality. *Similarity* ↑ indicates whether increasing scores imply increasing similarity of models.

Projection Weighted CCA. Morcos et al. [101] proposed Projection Weighted CCA (PWCCA) as an alternative to SVCCA. They argued that a representational similarity measure should weigh the individual canonical correlations ρ_i by their importance, i.e., the similarity of the *canonical variables* $\boldsymbol{R}\boldsymbol{w}_{\boldsymbol{R}}^{(i)}$ with the raw representation $\boldsymbol{R}$.

For that purpose, given mean-centered representations, they defined a weighting coefficient $\widetilde{\alpha}_i = \sum_{j=1}^{D} |\langle \boldsymbol{R}\boldsymbol{w}_{\boldsymbol{R}}^{(i)}, \boldsymbol{R}_{-,j} \rangle|$ for every canonical correlation ρ_i that models its importance. These coefficients are then normalized to weights $\alpha_i = \widetilde{\alpha}_i / \sum_j \widetilde{\alpha}_j$, yielding the final representational similarity measure

$$m_{\text{PWCCA}}(\boldsymbol{R}, \boldsymbol{R'}) = \textstyle\sum_{i=1}^{D} \alpha_i \rho_i. \tag{14}$$

This measure is asymmetric, since the weights α_i are only computed based on $\boldsymbol{R}$. Further, it is invariant to isotropic scaling and translation. PWCCA is bounded in the interval $[0, 1]$, with a value of one indicating equivalent representations.

3.2 Alignment-Based Measures

The next group of measures stipulates that a pair of representations $\boldsymbol{R}, \boldsymbol{R'}$ can be compared directly once the corresponding representation spaces have been aligned to each other. Alignment is usually realized by finding an optimal transformation $\varphi \in \mathcal{T}$ that minimizes a difference of the form $\|\varphi(\boldsymbol{R}) - \boldsymbol{R'}\|$. The exact group of transformations $\mathcal{T}$ used for alignment also directly determines and usually corresponds to the group of transformations that the corresponding

measure will be invariant to. Such direct alignment is only possible if the number of neurons in both representations are equal. Thus, we assume throughout the next section that $D = D'$, unless otherwise mentioned. We now discuss existing measures from this category.

Orthogonal Procrustes. The orthogonal Procrustes problem is a classical problem of finding the best orthogonal transformation to align two matrices in terms of minimizing the Frobenius norm (Eq. A.2) of the difference. Solving the problem leads to the similarity measure

$$m_{\text{Ortho-Proc}}(\boldsymbol{R}, \boldsymbol{R'}) = \min_{\boldsymbol{Q} \in \mathrm{O}(D)} \|\boldsymbol{RQ} - \boldsymbol{R'}\|_F = (\|\boldsymbol{R}\|_F^2 + \|\boldsymbol{R'}\|_F^2 - 2\|\boldsymbol{R}^\mathsf{T}\boldsymbol{R'}\|_*)^{\frac{1}{2}}, \tag{15}$$

where $\|\cdot\|_*$ denotes the nuclear norm (Eq. A.3) of a matrix [123]. The second formulation can also be used if $D \neq D'$ [65]. Ding et al. [34] used the square of $m_{\text{Ortho-Proc}}$ as a similarity score. By design, this measure is invariant to orthogonal transformations, and Williams et al. [155] showed that this measure satisfies the properties of a distance metric. This also holds when one optimizes Equation 15 over any subgroup $\mathrm{G}(D) \subset \mathrm{O}(D)$. Notably, considering the subgroup of permutation matrices yields the *Permutation Procrustes* measure [155], also known as *one-to-one matching distance* [65].

A similar optimization was proposed by Godfrey et al. [47] in their G_{ReLU}-Procrustes measure, which is designed to be invariant to $\mathrm{G}_{\mathrm{ReLU}}$ transformations, a special set of linear transformations (see Appendix A.4). Williams et al. [155] further proposed a variant that is invariant to spatial shifts in convolutional layers.

Generalized Shape Metrics. Williams et al. [155] applied theory of statistical shape analysis on the problem of measuring representational similarity. In that context, they also defined novel similarity measures. For representations with unit Frobenius norm (Eq. A.2) and any subgroup $\mathrm{G}(D) \subseteq \mathrm{O}(D)$, they introduced the *Angular Shape Metric*

$$m_\theta(\boldsymbol{R}, \boldsymbol{R'}) = \min_{\boldsymbol{Q} \in \mathrm{G}(D)} \arccos\langle \boldsymbol{RQ}, \boldsymbol{R'}\rangle_F, \tag{16}$$

which is invariant to transformations from $\mathrm{G}(D)$. To obtain a more general measure that is not restricted to representations preprocessed to unit norm, they apply the partial whitening function $\phi_\alpha(\boldsymbol{R}) = \boldsymbol{H}_N\boldsymbol{R}(\alpha\boldsymbol{I}_D + (1-\alpha)(\boldsymbol{R}^\mathsf{T}\boldsymbol{H}_N\boldsymbol{R})^{-1/2})$, where $\alpha \in [0, 1]$ and $\boldsymbol{H}_N = \boldsymbol{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\mathsf{T}$ denotes a centering matrix. This yields the *Partial Whitening Shape Metric*

$$m_{\theta,\alpha}(\boldsymbol{R}, \boldsymbol{R'}) = \min_{\boldsymbol{Q} \in \mathrm{O}(D)} \arccos \frac{\langle \phi_\alpha(\boldsymbol{R})\boldsymbol{Q}, \phi_\alpha(\boldsymbol{R'})\rangle_F}{\|\phi_\alpha(\boldsymbol{R})\|_F \|\phi_\alpha(\boldsymbol{R'})\|_F}. \tag{17}$$

For all $\alpha > 0$, this metric is invariant to orthogonal transformations and translations. For $\alpha = 1$, it is further invariant to isotropic scaling, for $\alpha = 0$ it is even invariant to affine transformations. Williams et al. [155] showed that this metric is also related to (regularized) canonical correlations. Both shape metrics are bounded in the interval $[0, \pi]$ and satisfy the properties of a distance metric.

Duong et al. [37] generalized the metrics from Williams et al. [155] to stochastic neural networks such as variational autoencoders [67], which map to distributions of representations instead of deterministic representations. Ostrow et al. [109] further extended this metric to measure similarity of dynamical systems, such as RNNs.

Soft Matching Distance. Khosla and Williams [65] generalized the Permutation Procrustes measure to settings in which the number of neurons in the representations $\boldsymbol{R}, \boldsymbol{R'}$ differ, i.e., $D \neq D'$. This was done by interpreting the problem of matching neurons as a transportation problem with possible solutions in the transportation polytope $\mathrm{TP}(D, D')$ [32]. Thus, assuming the representations are centered and scaled to unit norm, they defined the *soft matching distance* as

$$m_{\text{SoftMatch}}(\boldsymbol{R}, \boldsymbol{R'}) = \sqrt{\min_{\boldsymbol{P} \in \mathrm{TP}(D,D')} \sum_{ij} \boldsymbol{P}_{ij} \|\boldsymbol{R}_{-,i} - \boldsymbol{R'}_{-,j}\|_2^2}. \tag{18}$$

This measure is a special case of the 2-Wasserstein distance, and thus a metric [65]. Further, it is invariant to permutations, translations and scaling. Khosla and Williams [65] also proposed a variant that is related to Correlation Match (Eq. 21).

Linear Regression. An approach similar to Procrustes, but not restricted to orthogonal transformations, is based on predicting one representation from the other with a linear transformation [81, 73]. Then, the R-squared score of the optimal fit can be used to measure similarity [73]. Assuming mean-centered representations, this yields the measure

$$m_{R^2}(\boldsymbol{R}, \boldsymbol{R'}) = 1 - \frac{\min_{\boldsymbol{W} \in \mathbb{R}^{D \times D}} \|\boldsymbol{R'} - \boldsymbol{RW}\|_F^2}{\|\boldsymbol{R'}\|_F^2} = \frac{\left\|\left(\boldsymbol{R'}(\boldsymbol{R'}^\mathsf{T}\boldsymbol{R'})^{-1/2}\right)^\mathsf{T}\boldsymbol{R}\right\|_F^2}{\|\boldsymbol{R}\|_F^2}. \tag{19}$$

This asymmetric measure is invariant to orthogonal transformation and isotropic scaling. A value of one indicates maximal similarity, lower values indicate lower similarity. This measure has no lower bound.